

# The Burkholder Martingale Transform Inequality in Lean

## 1 Overview

This note describes the formal Lean theorem

`Lp_Burkholder_inequality_martingaleTransform`

proved in `MartingaleTransforms.lean`. The result is the real-valued martingale transform inequality

$$\|(w \star_m f)_n\|_{L^p(\mu)} \leq (p^* - 1) \|f_n\|_{L^p(\mu)}$$

for every time  $n$ , where  $1 < p < \infty$ ,  $f$  is a martingale, and  $w$  is a predictable multiplier bounded by 1 in absolute value.

In Lean the theorem is stated with the extended nonnegative exponent  $p : \mathbb{R}_{\geq 0, \infty}$ . The hypotheses include  $1 < p$ ,  $p \neq \infty$ , and the proof uses *p.toReal* for the real exponent in the Burkholder functions.

## 2 Basic Definitions

### 2.1 Martingale differences

For a discrete real-valued process  $f : \mathbb{N} \rightarrow \Omega \rightarrow \mathbb{R}$ , the martingale difference sequence is defined by

$$d_0 f = f_0, \quad d_{n+1} f = f_{n+1} - f_n.$$

In Lean this is

$$\text{martingaleDiff } f \ 0 = f \ 0, \quad \text{martingaleDiff } f \ (n+1) = f \ (n+1) - f \ n.$$

### 2.2 Martingale transforms

Given a multiplier process  $v : \mathbb{N} \rightarrow \Omega \rightarrow \mathbb{R}$ , the discrete martingale transform of  $f$  by  $v$  is

$$(v \star_m f)_n = \sum_{i=0}^n v_i d_i f.$$

The Lean definition is

`martingaleTransform v f := fun n => (Finset.range (n + 1)).sum fun i => v i * martingaleDiff`

The notation used in the file is

$$v \star_m f.$$

The increment identity proved in Lean is

$$(v \star_m f)_{n+1} - (v \star_m f)_n = v_{n+1}(f_{n+1} - f_n).$$

### 2.3 Predictability

The multiplier  $w$  is assumed to be strongly predictable with respect to a filtration  $\mathcal{F}$ :

$$\text{IsStronglyPredictable } \mathcal{F} w.$$

This is the formal version of the usual condition that  $w_{n+1}$  is  $\mathcal{F}_n$ -measurable. Predictability is what allows the proof to pull the multiplier out of conditional expectations when showing that the linear terms have expectation zero.

### 2.4 The auxiliary transforms $X$ and $Y$

The proof uses two transforms associated to  $w$ :

$$w_n^+ = w_n + 1, \quad w_n^- = w_n - 1.$$

In Lean these are `plusOne w` and `minusOne w`. The notation

$$X_n[w, f] = ((w + 1) \star_m f)_n, \quad Y_n[w, f] = ((w - 1) \star_m f)_n$$

is introduced by scoped notation. These satisfy the algebraic identities

$$\frac{X_n + Y_n}{2} = (w \star_m f)_n, \quad \frac{X_n - Y_n}{2} = f_n.$$

The final theorem uses these identities together with the definition of the Burkholder function  $v$ .

## 3 The Main Martingale Transform Theorem

The Lean theorem has the following mathematical content.

**Theorem 1** (Burkholder martingale transform inequality). *Let  $(\Omega, \mathcal{A}, \mu)$  be a finite measure space and let  $(\mathcal{F}_n)_{n \in \mathbb{N}}$  be a filtration. Let  $1 < p < \infty$ . Suppose  $f = (f_n)$  is a real-valued martingale,  $f_n \in L^p(\mu)$  for every  $n$ , and  $w = (w_n)$  is strongly predictable with*

$$|w_n(\omega)| \leq 1$$

*for every  $n$ , almost everywhere. Then for every  $n$ ,*

$$\|(w \star_m f)_n\|_{L^p(\mu)} \leq (p^* - 1) \|f_n\|_{L^p(\mu)},$$

*where*

$$q = \frac{p}{p-1}, \quad p^* = \max\{p, q\}.$$

In Lean, the constant is written as

$$\text{ENNReal.ofReal } (\text{Burkholder.pStar } p.\text{toReal} - 1).$$

## 4 The Burkholder Function $v$

The proof is organized around the Burkholder expression

$$v(p, x, y) = \left| \frac{x+y}{2} \right|^p - |p^* - 1|^p \left| \frac{x-y}{2} \right|^p.$$

This is the Lean definition `Majorants.v`. Substituting

$$x = X_n[w, f], \quad y = Y_n[w, f]$$

and using the identities above gives

$$v(p, X_n, Y_n) = |(w \star_m f)_n|^p - |p^* - 1|^p |f_n|^p.$$

Thus the inequality

$$\int v(p, X_n, Y_n) d\mu \leq 0$$

is exactly the  $p$ -power form of the martingale transform estimate. The last step of the Lean proof unfolds the definition of  $v$ , converts the integral inequality into an inequality of real  $L^p$ -norms, and then converts back to `eLpNorm`.

## 5 The Majorant Theorem exists\_majorant\_p\_g\_1

The analytic heart of the formalization is

$$\text{exists\_majorant\_p\_g\_1.}$$

For every real  $p > 1$ , it produces functions

$$u, \quad u_x, \quad u_y : \mathbb{R} \rightarrow \mathbb{R} \rightarrow \mathbb{R}$$

and a constant  $C \geq 0$  such that:

- $u, u_x, u_y$  are continuous on all of  $\mathbb{R}^2$ ;
- $u$  has  $p$ -growth:

$$|u(x, y)| \leq C(|x|^p + |y|^p);$$

- the selected derivatives have  $(p - 1)$ -growth:

$$|u_x(x, y)|, |u_y(x, y)| \leq C(|x|^{p-1} + |y|^{p-1});$$

- $u$  satisfies the tangent/concavity inequality in martingale-transform directions:

$$hk \leq 0 \implies u(x + h, y + k) \leq u(x, y) + u_x(x, y)h + u_y(x, y)k;$$

- $u$  majorizes  $v$ :

$$v(p, x, y) \leq u(x, y);$$

- $u$  is nonpositive on the region  $xy \leq 0$ :

$$xy \leq 0 \implies u(x, y) \leq 0.$$

### 5.1 Importancia do majorante $u$

O papel do majorante  $u$  é transformar uma desigualdade de norma em uma desigualdade de supermartingales. A função  $v$  contém diretamente a desigualdade desejada, mas ela não possui, por si só, a propriedade de concavidade adequada para aplicar a estrutura de martingales. O majorante  $u$  resolve isso:

1. Como  $v \leq u$ , basta provar que a esperança de  $u(X_n, Y_n)$  é não positiva.
2. A desigualdade tangente de  $u$  mostra que

$$u(X_{n+1}, Y_{n+1}) \leq u(X_n, Y_n) + u_x(X_n, Y_n)(X_{n+1} - X_n) + u_y(X_n, Y_n)(Y_{n+1} - Y_n).$$

3. A condição  $|w| \leq 1$  implica

$$(X_{n+1} - X_n)(Y_{n+1} - Y_n) \leq 0,$$

exatamente a hipótese necessária para a desigualdade tangente.

4. Pela previsibilidade de  $w$  e pela propriedade de martingale de  $f$ , os termos lineares têm integral zero.
5. Portanto,

$$\int u(X_{n+1}, Y_{n+1}) d\mu \leq \int u(X_n, Y_n) d\mu.$$

6. No tempo inicial,  $X_0 Y_0 \leq 0$ , então  $u(X_0, Y_0) \leq 0$ . Logo  $\int u(X_n, Y_n) d\mu \leq 0$  para todo  $n$ .

Combining this with  $v \leq u$  gives

$$\int v(p, X_n, Y_n) d\mu \leq 0,$$

and the definition of  $v$  then yields the desired martingale transform inequality.